

2023 U.S. Customer's Credit Card Data Analysis

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1 Objective

2023 U.S. Customer's Credit Card Data Analysis project designed to analyze and visualize 2023 U.S. credit card transaction data [1]. The goal was to uncover key trends, insights, and patterns in consumer spending behavior, with a specific focus on how credit card usage shapes broader financial transactions. According to the 2024 Diary of Consumer Payment Choice by the Federal Reserve Bank of Atlanta [2], credit cards account for 32% of payments, underscoring their significant role in consumer spending. By leveraging data visualization techniques, This project aimed to provide businesses and financial institutions with insights that can enhance customer engagement and financial decision-making.

2 Key Outcomes of the Analysis

2.1 Customer Behavior Analysis

- **Demographic Insights:** The analysis revealed that customers aged **40-50**, especially those in **Business** and **White Collar Jobs**, contribute the most to revenue. This demographic represents a high spending potential, making it an important segment for businesses to target.

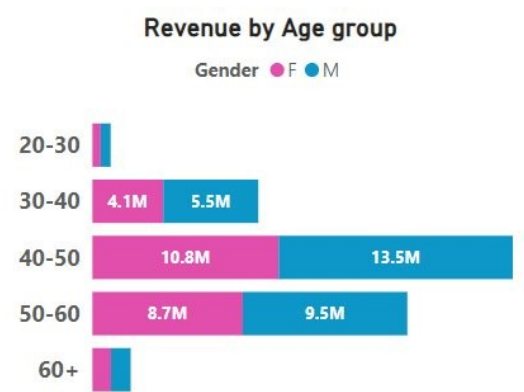


Figure 1: Generated Revenue by Age Group

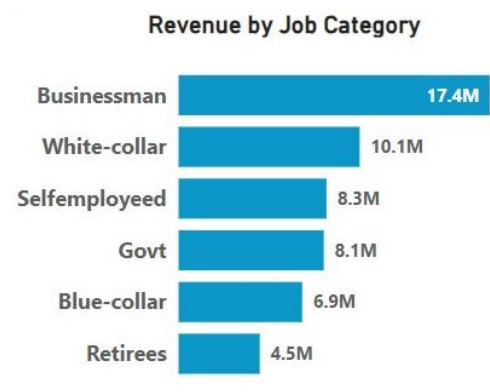


Figure 2: Revenue Generated by Job Category

- **Top Spending Categories:** **Bills** (25%) and **Entertainment** (17%) dominate consumer spending, suggesting that businesses in these sectors should focus their marketing strategies around these categories.
- **Gender-Based Spending:** **Male** customers generated approximately **\$30M** in revenue, while **Female** customers contributed **\$25M**. Interestingly, females in lower income categories significantly impacted revenue, while males in higher income groups were the dominant contributors.

2.2 Customer Satisfaction Insights

- **Satisfaction Scores:** The average satisfaction score across all customers was **3.19** (on scale of 1 - 5), indicating moderate satisfaction. The **Platinum** card category had the lowest score of **2.72**, signaling areas for improvement, possibly due to high fees or limited benefits compared to other card categories.

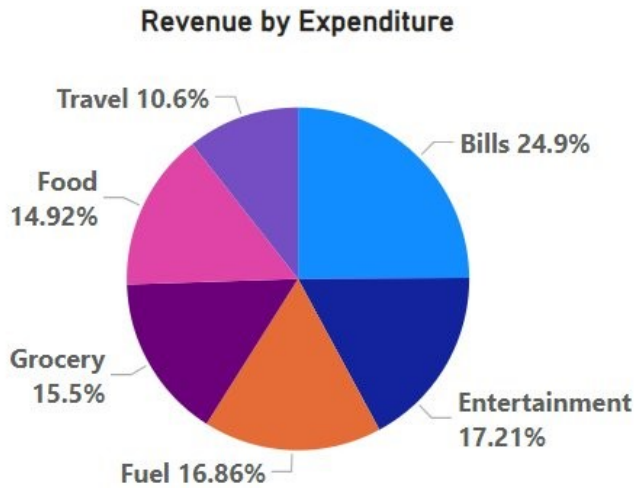


Figure 3: Revenue Generated by Expenditure Type



Figure 4: Gender Based Spending

- Low satisfaction levels in the Platinum category suggest that improving card benefits and customer service could help retain this customer segment.

Row	Card_Category	Customer_Satisfaction_Score	Total_Card_Holders
1	Silver	3.22	639
2	Blue	3.19	9214
3	Gold	3.04	188
4	Platinum	2.72	67

Figure 5: Customer Satisfaction Scores by Card Category

2.3 Financial Performance Insights

- Revenue Breakdown: Total revenue for 2023 was **\$55.3 million**, with the **Blue** card category contributing the largest share. **Q3** saw the highest revenue, while **Q4** underperformed, suggesting the need for further exploration of consumer behavior during the holiday season.
- Regional Revenue Distribution: The top 5 states — Texas, New York, California, Florida, and New Jersey—account for **94%** of total revenue, indicating that marketing efforts in these states should be prioritized.
- Payment Method Preferences: **Swipe** transactions generated **63%** of revenue, confirming that swipe payments remain the preferred method among cardholders.

2.4 Delinquency Insights

The analysis revealed a **6% delinquency rate**, indicating a portion of customers facing challenges in meeting payment obligations. This highlights an opportunity for improved risk management and targeted customer outreach strategies.

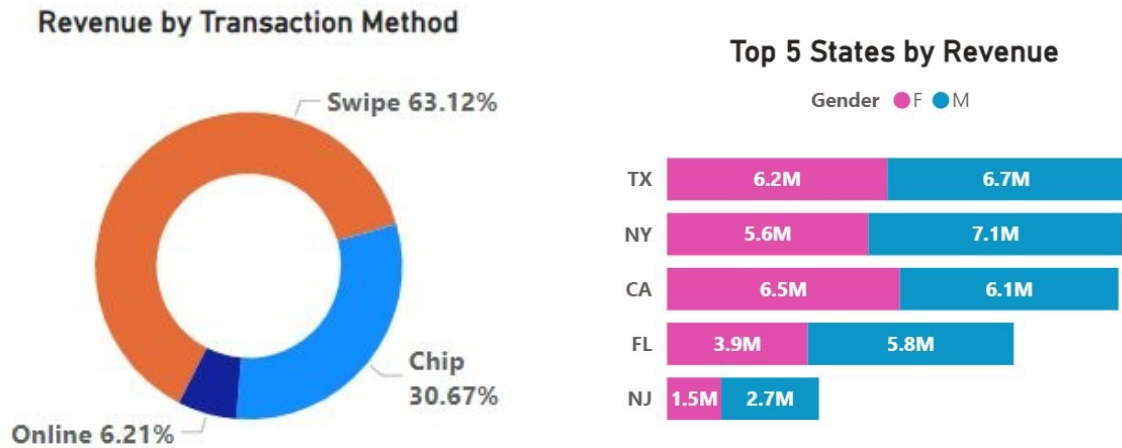


Figure 6: Revenue by Card Usage

Figure 7: Revenue by Top 5 States

Row	Quarter	revenue
1	Q3	14235479.69
2	Q1	13964362.77
3	Q2	13820566.72
4	Q4	13295001.05

Figure 8: Revenue Generated by Quarter

3 Possible Business Implications

The insights derived from this analysis can inform strategic decision-making across multiple business dimensions. Key implications include:

- **Targeted Marketing Campaigns:** Understanding spending trends across age groups, income brackets, and job categories allows businesses to tailor their marketing campaigns to specific customer segments. For example, focusing on customers aged 40-50 in high-revenue job categories such as Business and White Collar can maximize returns.
- **Revenue Optimization During Holidays:** Despite key holidays in Q4, revenue growth was at its lowest, highlighting the need for businesses to reassess their holiday marketing strategies and promotions. Tailored campaigns and incentives during this period could help capture untapped market potential and boost sales.
- **Product Development and Customization:** The findings reveal dissatisfaction with the Platinum card category due to potential issues like higher fees or fewer benefits. This insight can guide the development of more competitive products or the improvement of existing offerings to enhance customer satisfaction.
- **Localized Strategies for Revenue Growth:** The state-wise revenue analysis highlights regions such as Texas, New York, and California as high contributors. This data can guide resource allocation, branch expansion, and localized marketing initiatives to capitalize on these markets.

4 Tools & Technologies Used

- **PostgreSQL**: Used for storing and managing credit card transactional data.
- **pgAdmin**: Utilized for data cleaning, preprocessing, and initial analysis using SQL queries.
- **Power BI**: Employed for designing and visualizing three interactive real time dashboards [3].

5 Conclusion

This analysis provides a comprehensive view of consumer behavior, spending patterns, and transaction trends using 2023 U.S. credit card data. Key findings include the dominance of swipe transactions, high revenue contributions from specific age groups and income brackets, and the pivotal role of specific states in driving revenue. Additionally, the unexpected dip in Q4 revenue despite key holidays highlights opportunities to refine holiday sales strategies and promotions.

The dynamic dashboards [3] created in this project serve as valuable tools for monitoring trends, identifying opportunities, and addressing challenges. Businesses can leverage these insights to refine their strategies, improve customer satisfaction, and unlock untapped revenue potential. By bridging the gap between data and actionable intelligence, this project demonstrates the transformative power of data analytics in shaping a more customer-centric and competitive financial ecosystem.

References

- [1] Author. “US 2023 Credit Card Dataset”. In: *Kaggle* (2024). Accessed: December 01, 2024. URL: <https://www.kaggle.com/datasets/mrjashpatel/credit-card>.
- [2] Alex Miller. “The Most Popular Payment Methods in the U.S. [2024 Statistics & Data]”. In: *Upgraded Points* (2024). URL: <https://upgradedpoints.com/finance/most-popular-payment-methods-in-us/>.
- [3] Jash Patel. “Power BI Dashboards for Credit Card Data Analysis”. In: (2024). URL: https://github.com/dev-jash15/credit-card-financial-dashboard/blob/main/Report_Credit_Card_Dashboard.pdf.